

Robust learning-enabled voltage control for water–electricity systems under incomplete topology and parameter information

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HIGHLIGHTS

- Develops a unified learning-control framework for water–electricity systems under incomplete information.
- Enables online joint identification of grid topology and hydraulic parameters via nested convex body chasing.
- Designs robust MPC with a dynamic buffer for safe voltage and hydraulic-head regulation.
- Achieves reliable operation with only 0.0322 kV voltage violations under severe uncertainties.

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ABSTRACT

Water–electricity coupled systems play a vital role in the integrated management of energy and water resources, improving resource efficiency and system resilience. However, frequent topological switching, communication gaps, and time-varying electrical couplings severely challenge voltage stability. This paper develops a unified learning-control framework for water–electricity coupled systems under incomplete topology and parameter information, aiming to achieve robust voltage control while improving voltage stability and hydraulic safety. A nested convex body chasing algorithm enables joint online identification, while a robust model predictive controller with a dynamic buffer term compensates for estimation errors and disturbances. Theoretical guarantees are provided via finite bounds on constraint violations under worst-case conditions. The proposed method is validated on a coupled system consisting of a Southern California Edison 56-bus power distribution network and a 15-node water distribution network. Simulation results demonstrate that, even under the challenging case where both the power-grid topology and the WDN parameters are completely unknown, the proposed method largely maintains system-wide voltages within the prescribed safety limits. The average voltage-violation magnitude over the entire dispatch horizon is only 0.0322 kV, showing a clear improvement over the no control and two-stage control strategies. Meanwhile, the WDN experiences no hydraulic-head violations and retains an average hydraulic-head safety margin of 16.679 m.

1. Introduction

Water–electricity coupled systems are critical infrastructures that enable synergistic management of energy and water resources, improving regional resource efficiency and enhancing the resilience of both grids and water supply networks. With increasing integration of distributed energy resources (DERs), such as inverter-based generation,

energy storage, and flexible loads, opportunities for more sustainable and economical power systems continue to expand [1]. However, their intermittent and rapidly fluctuating outputs, often on timescales of seconds to minutes, challenge conventional voltage regulation, which relies on slow mechanical switching devices and faces operational limitations under frequent variations [2]. Moreover, high R/X ratios in distribution networks, stochastic renewable generation, and unpredictable loads

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Nomenclature

Symbols

$\mathcal{G}_e = (\mathcal{N}_e, \mathcal{E}_e)$, $\mathcal{G}_w = (\mathcal{N}_w, \mathcal{E}_w)$ Directed graphs of the PDN and WDN
 $v(t)$, $h(t)$ Nodal voltage vector of the PDN and hydraulic-head vector of the WDN
 $\Theta^* = [R^* \ X^*]$ True PDN topology and line-parameter matrix
 Φ^* True WDN pipe-resistance parameter matrix
 $\hat{\Theta}(t)$, $\hat{\Phi}(t)$ Online estimates of the PDN topology matrix and WDN parameter matrix
 $u_p(t)$, $u_q(t)$, $u_w(t)$ Active-power, reactive-power, and water-flow control variables

$P_{\text{pump}}(t)$, Π Pump power consumption and pump-to-grid-node mapping matrix
 $Q(t)$, $S_T(t)$, $V_T(t)$ Pipe-flow vector, tank charging/discharging flow, and tank water volume
 $\omega_v(t)$, $\omega_w(t)$ External disturbances on the grid side and water side
 η_v , η_w Disturbance bounds for the PDN and WDN models
 ρ_v , ρ_w Robust safety radii for PDN and WDN parameter-estimation errors
 ξ_v , ξ_w , k_v , k_w Slack variables and robust buffer terms for voltage and hydraulic-head constraints
 α_v , α_p , α_q , α_w , α_T , λ , β_v , β_w Weighting coefficients in the RMPC objective function

from electric vehicle charging contribute to voltage instability, resulting in frequent and persistent over- or under-voltage violations that threaten system security and pose severe challenges to the system's dynamic stability [3,4]. In water–electricity coupled systems, these voltage issues are further intertwined with hydraulic operation because pump scheduling directly affects power demand and, in turn, network voltage profiles.

In practical systems, accurate and up-to-date models are rarely available due to limited observability and frequent topology reconfiguration. As a result, control performance can degrade when topology is uncertain or line parameters vary over time, and similarly, when hydraulic parameters in the WDN are inaccurate. For power distribution networks (PDNs), utilities often possess only partial information regarding their primary and/or secondary networks and associated line impedances [5]. Even when the line infrastructure is known, the currently energized configuration may still be uncertain because switch status changes, service restoration actions, and network reconfiguration are not always promptly recorded in distribution management systems [6]. The inability to fully grasp topology information causes traditional control methods based on accurate topology to degrade or even fail. Therefore, the design of voltage control methods must be robust to incomplete topology information, enabling them to achieve good control performance even when topology information is missing. To address the above issues, data-driven methods for voltage control under incomplete grid information have developed rapidly in recent years. These methods avoid using inaccurate or even erroneous topology information for state estimation and voltage control, reducing the negative impact of model accuracy gaps on control effectiveness [7–10]. However, these data-driven methods typically require large amounts of historical operating data to achieve good performance; missing or noisy measurements may degrade the accuracy of model identification and subsequent control decisions [11].

There are also methods that separate topology estimation from control, first using data-driven methods to estimate topology and parameter information, and then performing voltage control based on the generated estimates [11–13]. Furthermore, for methods separating topology estimation and control, if the obtained topology is inaccurate, the controller consistently performing voltage control based on an inaccurate topology may lead to system instability. However, a unified framework that jointly performs online identification and coordinated control using only streaming measurements, without relying on offline calibration or large historical datasets, is still lacking.

Parallel challenges exist in WDNs, where pipeline parameters such as roughness coefficients are often uncertain due to aging, corrosion, scaling, or structural degradation [14]. Although network connectivity is usually known, accurate hydraulic models still require calibration, which remains challenging under limited sensing and time-varying operating conditions. Existing calibration approaches, including singular-value-decomposition-based methods and response-surface-based roughness calibration, are mainly developed for offline use and often rely

on sufficiently rich historical datasets [15,16]. More recent online hydraulic estimation approaches, such as nonlinear-state-observer-based roughness calibration and extended-Kalman-filter-based state estimation, improve adaptivity but still target the hydraulic subsystem itself rather than coupled voltage control [17,18]. Consequently, current WDN studies provide limited support for real-time coordinated voltage regulation when hydraulic parameters are also uncertain [19].

Recent studies have explored the coordinated operation of water–electricity coupled systems from different perspectives. For example, some studies regard pumps or water networks as adjustable electric loads [20], investigate uncertainty-aware coordination of water-energy hubs [21], develop joint design methods for renewable microgrids and water supply systems [1], and optimize the operation of energy-water microgrids using intelligent scheduling methods [22]. These studies demonstrate the value of coordinated water–electricity operation in reducing operating costs, improving operational flexibility, and enhancing system resilience. However, most existing studies rely on accurate distribution-network topology, line parameters, or calibrated hydraulic models. In practical operation, distribution-network topology may be incomplete or not updated in time because of switching operations, fault restoration, and network reconfiguration, while WDN pipe parameters may deviate from their nominal values due to aging, corrosion, and limited sensing measurements.

As the operational complexity of water–electricity coupled systems increases, learning-based methods have been introduced to improve real-time decision-making capability. For example, deep reinforcement learning has been used for islanded energy-water microgrid operation by treating the water supply system as a virtual energy-storage resource with operational flexibility [22]; machine-learning models have been developed for the economic dispatch of islanded water-energy microgrids to approximate complex optimization models and reduce online computational burden [23]; and data-driven methods have also been applied to economic dispatch and distributed energy management of grid-connected water-energy microgrids [24]. In broader integrated energy systems, online data-stream modeling [25], fast probabilistic energy-flow analysis [26], multi-agent reinforcement learning [27], coupled-feature extraction and multitask learning [28], deep reinforcement learning [29], and graph representation learning [30] have been used for modeling, forecasting, dispatch, and energy management. Nevertheless, these studies mainly focus on economic operation, load forecasting, energy-flow calculation, or dispatch acceleration, while closed-loop safety control under simultaneous uncertainty in distribution-network topology, line parameters, and WDN pipe parameters remains insufficiently addressed.

Consequently, a unified framework is still lacking for water–electricity coupled systems that can implement control under incomplete information while providing theoretical safety guarantees. The proposed unified learning-control framework synchronously performs grid-topology identification, WDN pipeline-parameter estimation, and robust

Table 1

Comparison between the proposed method and representative existing studies.

Reference	Sources of uncertainty (Grid topology / Line parameters / Pipeline resistance)	System domain (PDN / WDN / Coupled)	Solution methodology (Online / Robust / Provable)
[7]	× / ✓ / ×	✓ / × / ×	✓ / × / ×
[8]	✓ / × / ×	✓ / × / ×	✓ / × / ×
[9]	× / ✓ / ×	✓ / × / ×	✓ / ✓ / ×
[10]	× / ✓ / ×	✓ / × / ×	✓ / × / ×
[11]	× / ✓ / ×	✓ / × / ×	× / × / ×
[12]	× / ✓ / ×	✓ / × / ×	× / × / ×
[13]	✓ / × / ×	✓ / × / ×	× / × / ×
[17]	× / × / ✓	× / ✓ / ×	✓ / × / ×
[20]	× / × / ×	✓ / ✓ / ✓	× / ✓ / ✓
[31]	× / × / ×	✓ / ✓ / ✓	× / × / ×
[32]	× / × / ×	✓ / × / ×	✓ / × / ×
[33]	× / × / ×	✓ / × / ×	✓ / × / ×
[34]	× / × / ×	✓ / × / ×	✓ / ✓ / ×
[35]	× / ✓ / ×	✓ / × / ×	✓ / ✓ / ×
This work	✓ / ✓ / ✓	✓ / ✓ / ✓	✓ / ✓ / ✓

Note: ✓ indicates that the corresponding item is considered or included, whereas × indicates otherwise.

control decision-making within the same closed loop. It can estimate incomplete topology and parameter information online and directly embed the estimates into RMPC, thereby avoiding the adverse impact of estimation errors caused by the separation between identification and control in conventional two-stage frameworks, while enhancing voltage and hydraulic-head safety. To more clearly position the proposed method relative to representative existing studies, Table 1 summarizes the main differences in terms of uncertainty sources, system scope, and methodological properties.

1.1. Research gaps

Reliable voltage regulation in water–electricity coupled systems is essential for safe and interdependent operation. This task is critically challenged by concurrent uncertainties from both unknown or misidentified grid topology in PDN and uncalibrated or time-varying hydraulic parameters in WDN. Existing control and identification methods lack the ability to handle these coupled uncertainties in a unified, real-time, and theoretically assured manner. The literature specifically highlights three salient gaps:

- (1) Prior voltage control frameworks for water–electricity coupled systems typically assume that either the grid topology or WDN parameters are fully known, or at most, consider uncertainty in only one of the two systems. A unified control framework capable of simultaneously handling grid-topology uncertainty and WDN parameter uncertainty remains insufficiently developed. This limitation prevents safe and stable operation when information from both systems is partially or completely unavailable.
- (2) Existing identification and control strategies are typically implemented in separate stages. Topology estimation, parameter calibration, and control decision-making are often performed sequentially, and many data-driven methods rely heavily on offline training or large volumes of historical data. This separation weakens adaptability in dynamically changing operating conditions and may propagate estimation errors directly into the control loop. Thus, real-time joint identification methods that avoid offline calibration and large historical datasets are still needed.
- (3) Although robust or chance-constrained optimization has been applied to uncertain water–electricity operation, existing studies rarely provide rigorous finite-step safety guarantees under simultaneous topology uncertainty, hydraulic-parameter uncertainty, and external disturbances. In particular, finite worst-case

bounds on voltage and hydraulic head violations are still unavailable in the literature on coupled real-time control. As a result, the finite-step safety performance of coupled real-time control remains theoretically undercharacterized.

1.2. Contributions

To bridge these gaps, this paper proposes a unified online learning-based robust control framework for voltage regulation in water–electricity coupled systems under incomplete topology and parameter knowledge. The main contributions are threefold:

- (1) We develop a novel collaborative control framework that simultaneously accounts for both power grid topology uncertainty and water network pipeline parameter uncertainty. This addresses a key limitation of existing studies, which usually assume that one or both subsystems are fully known, and thus enables safe and stable operation under incomplete system information.
- (2) We introduce a data-driven nested convex body chasing (NCBC) algorithm for the joint real-time identification of grid topology and hydraulic parameters. This method synchronously refines compact uncertainty sets for the grid impedance matrix and pipeline resistance coefficients without requiring offline calibration or exact models. This enables adaptive learning of coupled system parameters under time-varying operating conditions.
- (3) We design an RMPC scheme enhanced with a dynamic robust buffer term to compensate for estimation errors and external disturbances. The proposed controller coordinates active and reactive power injections together with hydraulic control actions to maintain voltage and hydraulic head safety under coupled uncertainty. Finite upper bounds on voltage and hydraulic-head violations are further derived to provide worst-case safety guarantees.

The remainder of this paper is organized as follows. Section 2 develops the linearized mathematical models for both PDN and WDN. Section 3 details the proposed online learning-based robust control framework, including the NCBC algorithm and robust optimization. Section 4 provides a rigorous theoretical analysis of the system's safety guarantees. Section 5 presents the simulation results and a comprehensive comparative analysis. Finally, Section 6 concludes the paper.

2. Model

This section outlines the mathematical models for PDN and WDN, which serve as the basis for subsequent online learning and control design under incomplete system information. For PDN, a linearized voltage model is derived from the DistFlow equations. For WDN, a linear approximation of hydraulic dynamics is obtained by applying the Hazen–Williams formula with successive linear approximation (SLA). These formulations are detailed in the following subsections.

2.1. Power distribution network model

PDN is modeled as a radial system represented by a directed graph $\mathcal{G}_e = (\mathcal{N}_e, \mathcal{E}_e)$, where $\mathcal{N}_e = \{0, 1, 2, \dots, n_e\}$ denotes the set of buses (with bus 0 representing the substation), and $\mathcal{E}_e \subset \mathcal{N}_e \times \mathcal{N}_e$ describes the connectivity of the lines. Under a radial topology, each bus $j \in \mathcal{N}_e \setminus \{0\}$ has a unique parent bus. The relationship between power flow and voltage is described by the DistFlow model [36]. By neglecting line losses, the DistFlow model can be linearized as follows:

$$v = R^*(p_c + p_e) + X^*(q_c + q_e) + v_0 \mathbf{1}_{n_e} \quad (1)$$

where v_0 denotes the squared voltage magnitude at the substation; $\mathbf{1}_{n_e}$ is an all-ones vector of length n_e . The net active and reactive power injections at bus j can be decomposed into controllable and uncontrollable components: p_c and q_c represent the controllable parts, while p_e and q_e

represent the uncontrollable ones. Water pumps are treated as electrical loads and are included in p_e . Specifically, p_e can be further decomposed as $p_e(t) = p_e^0(t) - \Pi P_{\text{pump}}(t)$, where $p_e^0(t)$ denotes the uncontrollable net active-power injection excluding water pumps, $P_{\text{pump}}(t)$ denotes pump power consumption, and Π is the mapping matrix between water pumps and grid buses. The matrices R^* and $X^* \in \mathbb{S}_+^n$ are symmetric positive definite matrices determined by the topology and line parameters, with elements defined as:

$$R_{ij}^* = 2 \sum_{h,k \in \mathcal{P}_i \cap \mathcal{P}_j} r_{hk}, \quad X_{ij}^* = 2 \sum_{h,k \in \mathcal{P}_i \cap \mathcal{P}_j} x_{hk} \quad (2)$$

where $\mathcal{P}_i$ denotes the unique path from bus 0 to bus i . The set $\mathcal{P}_i \cap \mathcal{P}_j$ represents the overlapping portion of the paths from bus 0 to buses i and j . Since branch resistances and reactances satisfy $r_{ij}, x_{ij} > 0$, both R^* and X^* are positive definite matrices. Define the external voltage term excluding pump loads as $v_{\text{ext}}^0(t) = R^* p_e^0(t) + X^* q_e(t) + v_0 \mathbf{1}_{n_e}$. Substituting it into (1) yields

$$v(t) = R^* p_c(t) + X^* q_c(t) - R^* \Pi P_{\text{pump}}(t) + v_{\text{ext}}^0(t). \quad (3)$$

To formulate the voltage state-update relation, we define the control variables and disturbance term as follows. Let the active-power control variable be $u_p(t) = p_c(t+1) - p_c(t)$, the reactive-power control variable be $u_q(t) = q_c(t+1) - q_c(t)$, and the pump power increment be $\Delta P_{\text{pump}}(t) = P_{\text{pump}}(t+1) - P_{\text{pump}}(t)$. By integrating the active-power control effect, $u_p'(t) = u_p(t) - \Pi \Delta P_{\text{pump}}(t)$ denotes the net active-power control increment, and the control vector is further stacked as $u_v(t) = [u_p'(t); u_q(t)]$. The grid disturbance term is defined as $\omega_v(t) = v_{\text{ext}}^0(t+1) - v_{\text{ext}}^0(t)$, which captures voltage fluctuations caused by variations in uncontrollable loads, renewable generation, measurement errors, and other external factors. We assume that this disturbance is bounded, i.e., $\|\omega_v(t)\|_\infty \leq \eta_v$, where $\eta_v \in [0, \bar{\eta}_v]$ and the upper bound $\bar{\eta}_v$ is known. Denoting the unknown grid topology matrix as $\Theta^* = [R^* \ X^*]$, combining these definitions with (3) gives the voltage state-update equation:

$$v(t+1) = v(t) + \Theta^* u_v(t) + \omega_v(t). \quad (4)$$

2.2. Water distribution network model

WDN is represented by a directed graph $\mathcal{G}_w = (\mathcal{N}_w, \mathcal{E}_w)$, where $\mathcal{N}_w = \{1, 2, \dots, n_w\}$ denotes the set of nodes, n_w denoting the total number of nodes. The set of pipes $\mathcal{E}_w$ includes both ordinary pipes and pipes equipped with pumps. The hydraulic head at each node forms the vector $h \in \mathbb{R}^{n_w}$, and the pipe flow rates form the vector $Q \in \mathbb{R}^{|\mathcal{E}_w|}$. Due to wall friction, head loss occurs along each pipe. For ordinary pipes, this loss is commonly described by the Hazen–Williams formula, which captures the nonlinear dependence of head loss on flow rate and pipe resistance. For an ordinary pipe connecting nodes i and j , the head loss can be expressed as:

$$h_i - h_j = \kappa_{ij} \cdot Q_{ij} |Q_{ij}|^{\beta-1} \quad (5)$$

where h_i and h_j denote the hydraulic heads at nodes i and j , respectively; κ_{ij} denotes the resistance coefficient of pipe (i, j) ; Q_{ij} represents the flow rate in pipe (i, j) ; and β indicates the exponent in the Hazen–Williams formula, reflecting the nonlinear effect of turbulent friction.

To facilitate online learning and convex optimization, the nonlinear head–flow relation must be linearized. This is achieved using SLA. By choosing an operating flow rate $Q_{ij,0}$, and performing a first-order Taylor expansion, the following linearized expression is obtained:

$$h_{ij} \approx \kappa_{ij} Q_{ij,0}^\beta (1 - \beta) + \kappa_{ij} \beta Q_{ij,0}^{\beta-1} Q_{ij} \quad (6)$$

Converting the above equation into a linear approximation model with respect to the flow rate variation $\Delta Q_{ij} = Q_{ij} - Q_{ij,0}$ gives:

$$\Delta h_{ij} = M(Q_{ij,0}) \cdot \Phi \cdot \Delta Q_{ij} \quad (7)$$

where $M(Q_{ij,0})$ is a time-varying diagonal matrix determined by the operating point flow rate, and Φ is a constant diagonal matrix whose diagonal elements represent the resistance coefficients of each pipe.

For pipes equipped with pumps, the head characteristics are determined by the pump head–flow curve. For a pump installed on pipe (i, j) , the relationship between the pump head h_{ij}^{pump} and flow rate Q_{ij} can be expressed as:

$$h_{ij}^{\text{pump}} = h_{0,ij} - S_{ij} Q_{ij}^\gamma \quad (8)$$

where h_{ij}^{pump} is the head gain between nodes i and j ; $h_{0,ij}$ denotes the pump shutoff head, representing the maximum head at zero flow rate; S_{ij} represents the pump characteristic coefficient; and γ indicates the pump curve exponent, typically $\gamma = 2$. The pump parameters $h_{0,ij}$, S_{ij} , and γ are assumed to be known design parameters obtainable from manufacturer specifications. Under normal pumping conditions, the flow rate is constrained to $0 \leq Q_{ij} \leq (h_{0,ij}/S_{ij})^{1/\gamma}$, ensuring $h_{ij}^{\text{pump}} \geq 0$. Linearizing the pump head equation at the operating point $Q_{ij,0}$ yields:

$$\Delta h_{ij}^{\text{pump}} \approx -S_{ij} \gamma Q_{ij,0}^{\gamma-1} \Delta Q_{ij} = M^{\text{pump}}(Q_{ij,0}) \Delta Q_{ij} \quad (9)$$

where $M^{\text{pump}}(Q_{ij,0})$ is a time-varying diagonal matrix with diagonal elements $-S_{ij} \gamma Q_{ij,0}^{\gamma-1}$ for pipes with pumps and zero for ordinary pipes.

In a WDN that includes both pump-induced head gains and pipe friction losses, the nodal head variation should be jointly determined by all pump head gains and pipe friction losses along the path from the water source to each node. The WDN considered in this paper is radial; therefore, there exists a unique hydraulic path from the source node to any other node, and all pipes are oriented from the water source toward downstream nodes. We introduce the path matrix $T \in \mathbb{R}^{n_w \times |\mathcal{E}_w|}$, where $T_{i,e} = 1$ if pipe e lies on the unique path from the water source to node i , and $T_{i,e} = 0$ otherwise. Hence, the nodal head variation can be expressed as

$$\Delta h_{\text{node}} = T \Delta h_{ij}^{\text{pump}} - T \Delta h_{ij}. \quad (10)$$

To obtain a formulation suitable for online parameter learning, substituting (7) and (9) into (10) and reorganizing the equation gives

$$T \Phi M(Q_{ij,0}) \Delta Q = T M^{\text{pump}}(Q_{ij,0}) \Delta Q - \Delta h_{\text{node}}. \quad (11)$$

This equation serves as the parameter learning formulation for WDN. Here, Δh_{node} represents the measured nodal head variation from time t to $t+1$. The left-hand side is caused by pipe friction losses, whereas the first term on the right-hand side is caused by pump head gains.

We further express the model in state-update form. Let the water-network control input be $u_w(t)$, which represents the feasible flow adjustment induced by pump operation, and introduce the disturbance term $\omega_w(t)$ to account for unmodeled hydraulic dynamics, water-demand fluctuations, linearization errors, and other external factors. This disturbance is assumed to satisfy $\|\omega_w(t)\|_\infty \leq \eta_w$, where $\eta_w \in [0, \bar{\eta}_w]$ and the upper bound $\bar{\eta}_w$ is known. Then, the WDN state-update equation can be written as

$$h(t+1) = h(t) + B^* u_w(t) + \omega_w(t), \quad (12)$$

where $B^* = T(M^{\text{pump}}(Q_{ij,0}) - \Phi^* M(Q_{ij,0}))$ denotes the hydraulic sensitivity matrix of the WDN. Here, t denotes a generic time step; different time intervals are used for the PDN and WDN.

To ensure basic hydraulic feasibility of pipe flows, nodal flow conservation is considered. Let $C \in \mathbb{R}^{(n_w-1) \times |\mathcal{E}_w|}$ denote the node-pipe incidence matrix excluding the source node, and let $d(t)$ denote the water-demand vector. Further considering the set of storage tanks $\mathcal{T}$ in the WDN, let $S_T(t)$ denote the net tank-charging flow vector, where $S_T(t) > 0$ represents tank charging and $S_T(t) < 0$ represents tank discharging. Let E_T denote the mapping matrix between storage tanks and WDN nodes. The nodal flow conservation relation is then given by

$$CQ(t) = d(t) + E_T S_T(t). \quad (13)$$

In addition, the storage tank state satisfies the mass conservation relation

$$V_T(t+1) = V_T(t) + S_T(t)\Delta t, \quad (14)$$

where $V_T(t)$ denotes the stored water volume of the tanks and Δt is the WDN dispatch time step. To ensure the physical feasibility of storage tanks, their volumes and charging/discharging rates satisfy

$$\underline{V}_T \leq V_T(t+1) \leq \bar{V}_T, \quad (15)$$

$$-\bar{S}_T^{\text{dis}} \leq S_T(t) \leq \bar{S}_T^{\text{ch}}, \quad (16)$$

where $\underline{V}_T$ and $\bar{V}_T$ denote the lower and upper bounds of tank volume, respectively, and $\bar{S}_T^{\text{dis}}$ and $\bar{S}_T^{\text{ch}}$ denote the maximum discharging and charging rates of the tanks, respectively. It should be noted that storage tanks are modeled as controllable nodal charging/discharging units in this paper, their influence on nodal hydraulic heads is captured through the nodal flow conservation constraint and the resulting redistribution of pipe flows.

PDN and WDN are coupled through pump power consumption P_{pump} , which is a function of pipe flow rate Q_{ij} :

$$P_{\text{pump}}(Q_{ij}) = \frac{\rho g}{\eta_p} h_{ij}^{\text{pump}} Q_{ij} \quad (17)$$

where ρ is the water density, $g = 9.81 \text{ m/s}^2$ is the gravitational acceleration, η_p is the pump efficiency, and h_{ij}^{pump} is the pump head. To maintain convexity in the optimization, we define the constant $\zeta = \frac{\rho g}{\eta_p}$ and perform a first-order Taylor expansion at the operating point $Q_{ij,0}$ to obtain its linear upper bound:

$$P_{\text{pump}}(Q_{ij}) \leq a_{ij}^p Q_{ij} + b_{ij}^p \quad (18)$$

$$\begin{cases} a_{ij}^p = \zeta [h_{0,ij} - S_{ij}(\gamma + 1)Q_0^\gamma], \\ b_{ij}^p = \zeta S_{ij}\gamma Q_0^{\gamma+1}. \end{cases} \quad (19)$$

It should be noted that SLA does not assume that the linearized model is exact. The linearization error generated by the WDN equations is included in the water-network disturbance term $\omega_w(t)$, whereas the impact of the deviation between the linearized pump power and the actual pump power on grid-side active loads is included in the grid disturbance term $\omega_v(t)$. These effects are further compensated for by the robust buffer terms in RMPC.

Through SLA, the nonlinear hydraulic relations are reformulated into a convex matrix representation suitable for parameter learning. This modeling framework provides the mathematical basis for the NCBC algorithm introduced in Section 3.

2.3. Assumptions

To clarify the applicability of the proposed method, the following assumptions are made:

- (1) The true grid topology matrix Θ^* and water-network parameter matrix Φ^* belong to known compact convex uncertainty sets $\mathcal{X}_v$ and $\mathcal{X}_w$, respectively. This means that although the exact topology or parameters are unknown, their physically feasible ranges are available. The grid-side disturbance $\omega_v(t)$ and water-side disturbance $\omega_w(t)$ are bounded, satisfying $\|\omega_v(t)\|_\infty \leq \eta_v$ and $\|\omega_w(t)\|_\infty \leq \eta_w$, respectively.
- (2) Real-time measurements are available for online updating. Specifically, the controller can obtain measurements of nodal voltages, hydraulic heads, control inputs, and flow variations at each dispatch interval, which are used to update the historical dataset and shrink the parameter feasible sets.

3. Online learning-based robust control for water–electricity coupled systems

To address the challenges of incomplete topology and parameter information in coupled PDN and WDN, this section presents an online learning-based robust control framework. The proposed method performs joint real-time estimation of the grid impedance matrix and pipe resistance parameters, enabling coordinated voltage and hydraulic control under partial system knowledge.

3.1. Overview of algorithm framework

The algorithm integrates a NCBC mechanism for online parameter learning within an RMPC framework. Key unknown parameters are continuously updated from streaming measurement data, while the controller iteratively solves a robust optimization problem using the latest parameter estimates. This coupled learning-control design ensures operational safety under estimation inaccuracies and external disturbances.

The online learning-based robust control framework for the water–electricity coupled systems consists of two layers: an information learning layer and a control decision layer. The learning layer employs the NCBC algorithm to sequentially estimate the grid impedance matrix and the pipe resistance matrix from real-time measurements. Based on these estimates, the decision layer computes coordinated control actions through an RMPC formulation. The overall architecture of the online learning-based robust control framework for the water–electricity coupled systems is illustrated in Fig. 1.

3.2. Topology and parameter estimation based on NCBC

The NCBC algorithm forms the core of the information learning layer. It reformulates parameter learning as the problem of tracking a time-varying optimal point over nested convex sets. By sequentially integrating new measurements, NCBC progressively tightens the uncertainty sets of the grid topology and pipe parameters, steering the estimates toward their true values.

Define the parameter space for power grid topology as $\mathcal{X}_v = \mathcal{X}_R \times \mathcal{X}_X$ denoting a compact convex set for the impedance matrix. We assume that both the true topology matrix Θ^* and the estimate $\hat{\Theta}$ at each time step belong to a known compact convex set $\mathcal{X}_v$. At time step t , the feasible set P_t^{grid} is defined as the set of all topology parameters consistent with historical observations. Let the voltage prediction residual be $r_v(\tau) = v(\tau) - v(\tau-1) - \Theta u_v(\tau-1)$. The feasible set is:

$$P_t^{\text{grid}} = \left\{ (R, X, \eta_v) \left| \begin{array}{l} \|r_v(\tau)\|_\infty \leq \eta_v, \\ v(\tau) - \hat{\Theta} u_v(\tau-1) \in \mathcal{V}_{\text{ext}}, \\ \forall \tau \leq t-1 \end{array} \right. \right\} \quad (20)$$

where $\mathcal{V}_{\text{ext}}$ denotes the bounded set of PDN external disturbances.

The NCBC algorithm updates the convex feasible set by recursively tightening it as new measurements become available. Each observation introduces an additional linear constraint, and the updated feasible set is constructed as the intersection of the previous set with the new constraint:

$$P_{t+1}^{\text{grid}} = P_t^{\text{grid}} \cap \{ \|r_v(t+1)\|_\infty \leq \eta_v \}. \quad (21)$$

Once the latest feasible set P_{t+1}^{grid} is obtained, the NCBC algorithm computes the revised power grid topology estimate by solving the following projection problem. Let $\theta_v = (\hat{\Theta}_v, \hat{\eta}_v)$ denote the combined parameter vector. The estimate is then updated as:

$$\theta_v(t+1) = \arg \min_{\theta_v \in P_{t+1}^{\text{grid}}} \|\theta_v - \theta_v(t)\|_{\Delta, \delta_v} \quad (22)$$

where $\|\cdot\|_{\Delta, \delta}$ denotes the joint norm. For a symmetric matrix $A \in \mathbb{S}^n$, the triangular norm is defined as $\|A\|_\Delta = \|\text{vech}(A)\|_2$, where $\text{vech}(A)$ vectorizes the lower triangular part of A including the diagonal. For the parameter $\theta = (A, \eta)$ with weighting parameter $\delta > 0$, the joint norm is $\|\theta\|_{\Delta, \delta} =$

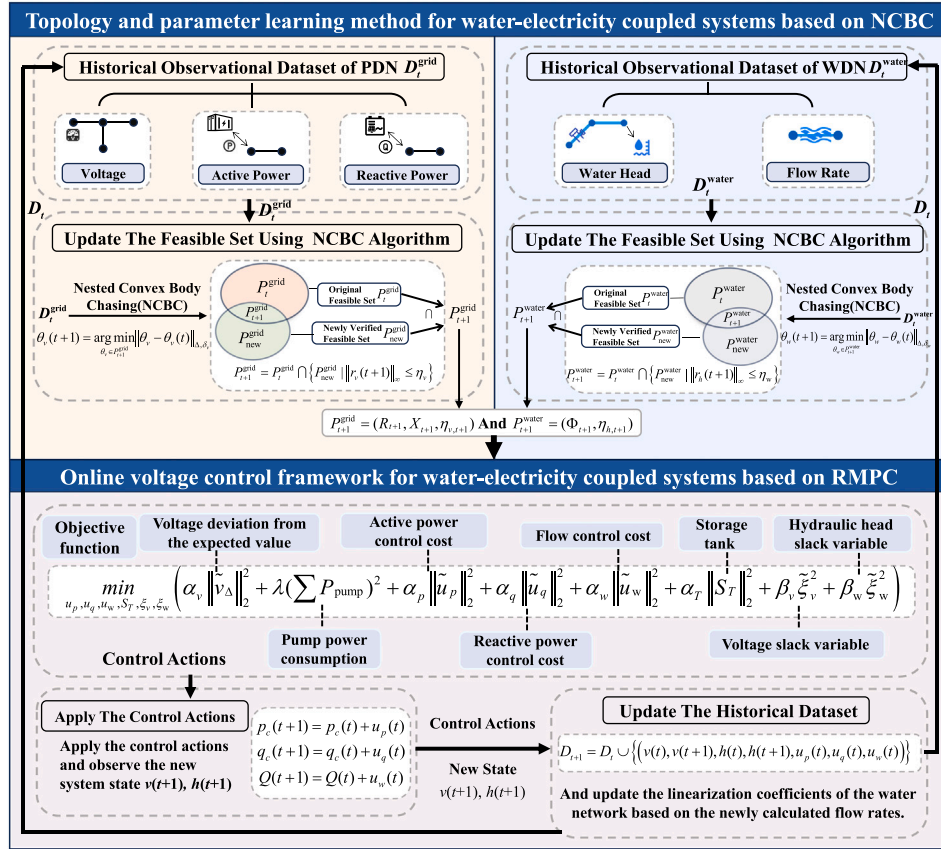


Fig. 1. Online learning-based robust control framework for water-electricity coupled systems.

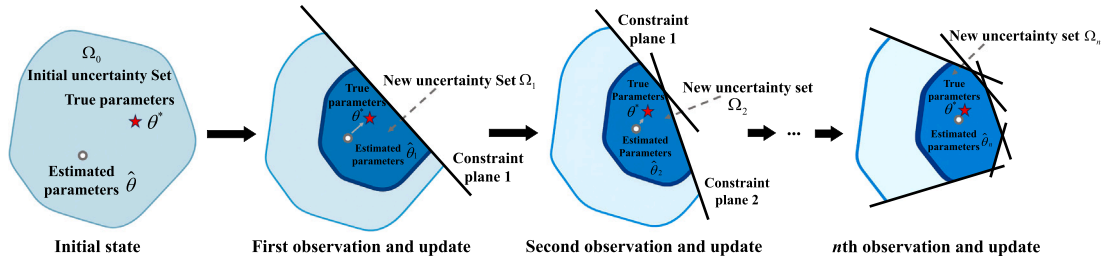


Fig. 2. NCBC algorithm for parameter learning.

$\sqrt{\|A\|_{\Delta}^2 + \delta^2 \eta^2}$. For the PDN matrix $\Theta = [R \ X]$, where R and X are symmetric matrices, we define $\|\Theta\|_{\Delta} = \sqrt{\|R\|_{\Delta}^2 + \|X\|_{\Delta}^2}$. Accordingly, the joint norm of $\theta_v = (R, X, \eta_v)$ is $\|\theta_v\|_{\Delta, \delta_v} = \sqrt{\|R\|_{\Delta}^2 + \|X\|_{\Delta}^2 + \delta_v^2 \eta_v^2}$. Eq. (22) determines the optimal estimate $\theta_v(t)$ within the current feasible set P_t^{grid} , starting from the prior estimate $\theta_v(t-1)$. By iteratively reducing the feasible set as new observations are incorporated, the topology estimate progressively approaches the true value.

Similarly, we define the parameter space for WDN as $\mathcal{X}_w = \mathcal{X}_{\Phi}$, representing the compact convex set for pipe resistance coefficients. We assume that both the true parameter matrix Φ^* and the predicted matrix $\hat{\Phi}$ belong to a known compact convex set $\mathcal{X}_{\Phi}$. Denoting the hydraulic head prediction residual as $r_h(\tau) = h(\tau) - h(\tau-1) - Bu_w(\tau-1)$, the feasible set at time step $t-1$ is:

$$P_t^{\text{water}} = \left\{ (\Phi, \eta_w) \mid \begin{array}{l} \|r_h(\tau)\|_{\infty} \leq \eta_w, \\ h(\tau) - Bu_w(\tau-1) \in \mathcal{H}_{\text{ext}}, \\ \forall \tau \leq t-1 \end{array} \right\} \quad (23)$$

where $\mathcal{H}_{\text{ext}}$ denotes the bounded set of WDN external disturbances. Similar to PDN, the feasible set of WDN is updated as:

$$P_{t+1}^{\text{water}} = P_t^{\text{water}} \cap \{\theta_w(t+1) \mid \|r_h(t+1)\|_{\infty} \leq \eta_w\} \quad (24)$$

Let $\theta_w = (\hat{\Phi}, \hat{\eta}_w)$ denote the combined parameter vector for WDN. The corresponding estimation update follows:

$$\theta_w(t+1) = \arg \min_{\theta_w \in P_{t+1}^{\text{water}}} \|\theta_w - \theta_w(t)\|_{\Delta, \delta_w} \quad (25)$$

By iteratively tightening the feasible set for water network parameters, the estimated pipe resistance matrix progressively converges toward its true value. The NCBC workflow is illustrated in Fig. 2.

The NCBC algorithm achieves a competitive ratio of $\gamma(m)$ in m -dimensional Euclidean space, satisfying:

$$\sum_{t=1}^T \|\theta_{t+1} - \theta_t\| \leq \gamma(m) \cdot \text{diam}(\mathcal{X}) \quad (26)$$

For all nested convex body sequences $X_1 \supseteq X_2 \supseteq \dots \supseteq X_T$. Here, $\text{diam}(\mathcal{X}) = \max_{\theta_1, \theta_2 \in \mathcal{X}} \|\theta_1 - \theta_2\|_{\Delta, \delta}$ denotes the diameter of the compact set $\mathcal{X} \in \mathbb{S}^n \times \mathbb{R}$, and θ represents the power grid parameters $\theta_v = (\Theta, \eta_v)$ or the water network parameters $\theta_w = (\Phi, \eta_w)$.

The competitive ratio of NCBC characterizes the efficiency of the optimization process in progressively approximating the true topology and parameters. For the projection-based NCBC algorithm, the competitive ratio is given by $\gamma(m) = \pi(m-1)m^{m/2}$. The dimension of the power grid parameter space satisfies $m_v = n_e(n_e + 1) + 1$, while that of the water network parameter space is $m_w = n_w$. The competitive ratio directly affects the convergence rate of parameter estimation and the overall control performance.

3.3. Robust optimal control for water–electricity coupled systems

After obtaining the topology and parameter estimates $\hat{\Theta}(t)$, $\hat{\eta}_v(t)$, $\hat{\Phi}(t)$, and $\hat{\eta}_w(t)$ through the NCBC algorithm, the controller solves the following robust optimization:

$$\min_{u_p, u_q, u_w, \bar{S}_T, \xi_v, \xi_w} \alpha_v \|\bar{v}_\Delta\|_2^2 + \lambda \left(\sum \bar{P}_{\text{pump}} \right)^2 + \alpha_p \|\bar{u}_p\|_2^2 + \alpha_q \|\bar{u}_q\|_2^2 + \alpha_w \|\bar{u}_w\|_2^2 + \alpha_T \|\bar{S}_T\|_2^2 + \beta_v \bar{\xi}_v^2 + \beta_w \bar{\xi}_w^2 \quad (27)$$

where $\alpha_v, \alpha_p, \alpha_q, \alpha_w, \alpha_T$ and λ are weighting coefficients for the individual sub-objectives, β_v and β_w are penalty weights for the slack variables. Since all terms are normalized, the weights represent the priorities of different operational objectives and can be used to balance safety and economic performance. In engineering applications, these weights can be dynamically adjusted according to operating conditions. Increasing α_v or β_v prioritizes voltage safety and helps avoid voltage constraint violations; increasing β_w enhances hydraulic-head safety; increasing $\alpha_p, \alpha_q, \alpha_w$ or α_T suppresses frequent regulation and reduces equipment wear; when both voltage and hydraulic head have sufficient safety margins, increasing λ can further reduce pump power consumption. The individual terms in the objective function possess distinct physical meanings and units, and summing them directly would introduce scaling disparities. To balance the contributions of all objectives, each term is normalized as follows:

$$\bar{v}_\Delta = \frac{\hat{v} - v_{\text{nom}}}{\bar{v} - \underline{v}}, \quad \bar{P}_{\text{pump}} = \frac{P_{\text{pump}}}{P_{\text{ref}}} \quad (28)$$

$$\bar{\xi}_v = \frac{\xi_v}{\bar{v} - \underline{v}}, \quad \bar{\xi}_w = \frac{\xi_w}{\bar{h} - \underline{h}} \quad (29)$$

$$\bar{u}_p = \frac{u_p}{\bar{u}_p - \underline{u}_p}, \quad \bar{u}_q = \frac{u_q}{\bar{u}_q - \underline{u}_q}, \quad \bar{u}_w = \frac{u_w}{\bar{u}_w - \underline{u}_w}, \quad \bar{S}_T = \frac{S_T}{S_T^{\text{ch}}} \quad (30)$$

where $\bar{u}_p - \underline{u}_p, \bar{u}_q - \underline{u}_q$, and $\bar{u}_w - \underline{u}_w$ denote the adjustable ranges of active power control, reactive power control and water network flow control, respectively; S_T^{ch} denotes the maximum tank-charging rate; and P_{ref} denotes the reference pump power.

Eq. (27) defines the objective function for the robust optimal control of the water–electricity coupled system. It consists of four weighted components aimed at minimizing voltage deviations from the nominal profile, reducing total pump energy consumption, limiting the magnitude of control actions, and penalizing constraint violations through slack variables. The optimization is performed subject to the following constraints:

$$\hat{v}(t+1) = v(t) + \hat{\Theta}(t)u_v(t) \quad (31a)$$

$$\hat{h}(t+1) = h(t) + \hat{B}(t)u_w(t) \quad (31b)$$

$$\underline{u}_p \leq u_p \leq \bar{u}_p, \quad \underline{u}_q \leq u_q \leq \bar{u}_q, \quad \underline{u}_w \leq u_w \leq \bar{u}_w \quad (31c)$$

$$\xi_v \geq 0, \quad \xi_w \geq 0 \quad (31d)$$

$$\underline{v}\mathbf{1} + (k_v - \xi_v)\mathbf{1} \leq \hat{v}(t+1) \leq \bar{v}\mathbf{1} - (k_v - \xi_v)\mathbf{1} \quad (31e)$$

$$\underline{h}\mathbf{1} + (k_w - \xi_w)\mathbf{1} \leq \hat{h}(t+1) \leq \bar{h}\mathbf{1} - (k_w - \xi_w)\mathbf{1} \quad (31f)$$

where constraints (31a) and (31b) represent the prediction models for PDN and WDN, respectively, using the estimated parameters to predict the system state at the next time step. Constraint (31c) specifies the upper and lower bounds on control variables. Constraint (31d) ensures the non-negativity of slack variables. Constraints (31e) and (31f) enforce the robust safety constraints for PDN and WDN, confining the predicted voltage and hydraulic head within specified ranges. In addition, the optimization problem should satisfy the WDN and storage-tank constraints presented in Section 2.2. Here, k_v and k_w are robust buffer terms, dynamic safety margins reserved inside the voltage and hydraulic-head constraint boundaries. They ensure that the actual system states can still satisfy safety constraints even in the presence of estimation errors and external disturbances. When control actions are large or model uncertainty is strong, the robust buffer terms increase accordingly. As NCBC progressively shrinks the parameter feasible sets and model uncertainty decreases, the buffer terms generally become smaller.

Once the optimization problem is solved, the optimal control actions $u_p(t)$, $u_q(t)$, and $u_w(t)$ are obtained. These control actions are then applied to the physical system to update the controllable active power $p_c(t+1) = p_c(t) + u_p(t)$, reactive power injections $q_c(t+1) = q_c(t) + u_q(t)$, and pipe flow rates $Q(t+1) = Q(t) + u_w(t)$.

After implementing the control actions, the resulting system states, namely the grid voltage $v(t+1)$ and nodal hydraulic head $h(t+1)$ are measured. The newly generated observation data from PDN ($v(t)$, $v(t+1)$, $u_p(t)$, $u_q(t)$) and WDN ($h(t)$, $h(t+1)$, $Q(t)$) are appended to the historical dataset for online learning of power grid topology and water network parameters at the next time step. Simultaneously, the linearization coefficients for WDN are updated. That is, for each pump $k \in \mathcal{P}$, the linearization coefficients a_k^p and b_k^p are recalculated based on the updated flow rates, ensuring an accurate linear representation for the subsequent control iteration.

4. Theoretical guarantees for system performance

This section provides rigorous theoretical guarantees for the proposed online learning-based robust control strategy under unknown power grid topology and water network parameters. In contrast to prior studies that primarily address single systems, this work develops a unified performance analysis framework tailored to water–electricity coupled systems. We derive joint finite-error bounds for the coupled dynamics and establish provable safety despite uncertainties in grid topology and hydraulic parameters.

Assume there exist compact convex sets $\mathcal{V}_{\text{ext}} \subset \mathbb{R}^{n_v}$ and $\mathcal{H}_{\text{ext}} \subset \mathbb{R}^{n_w}$, along with positive constants $\epsilon_v, \epsilon_w > 0$, such that for all $v_{\text{ext}} \in \mathcal{V}_{\text{ext}}$ and $h_{\text{ext}} \in \mathcal{H}_{\text{ext}}$, there exist feasible controls $u_v \in [\underline{u}_v, \bar{u}_v]$ and $u_w \in [\underline{u}_w, \bar{u}_w]$ satisfying:

$$\Theta^* u_v(t) + v_{\text{ext}} \in [\underline{v} + (\bar{\eta}_v + \epsilon_v)\mathbf{1}, \bar{v} - (\bar{\eta}_v + \epsilon_v)\mathbf{1}] \quad (32)$$

$$B^* u_w(t) + h_{\text{ext}} \in [\underline{h} + (\bar{\eta}_w + \epsilon_w)\mathbf{1}, \bar{h} - (\bar{\eta}_w + \epsilon_w)\mathbf{1}] \quad (33)$$

The above assumption represents offline, static control margins that ensure theoretical feasibility of the coupled system. This assumption does not rely on online estimation but follows from worst-case analysis. It essentially requires that the available control equipment capacity is sufficiently large to maintain safe operation even when worst-case disturbances and maximum parameter errors occur simultaneously.

To systematically handle uncertainty in the water–electricity coupled systems, we define compact parameter sets $\mathcal{X}_v$ and $\mathcal{X}_w$ containing all possible power grid topology matrices Θ and water network parameter matrices Φ . Their diameters $D_v = \text{diam}(\mathcal{X}_v)$ and $D_w = \text{diam}(\mathcal{X}_w)$ quantify the magnitude of the initial uncertainty in power grid topology and water network parameters. The objective of the online learning algorithm is to continually refine these parameter sets by processing streaming input–output data, thereby shrinking the uncertainty and improving model accuracy over time.

In the information learning layer, the NCBC algorithm progressively refines the estimates of grid topology and water network parameters by recursively intersecting the feasible sets that are consistent with historically observed data in the real system. At each time step, the NCBC algorithm integrates the latest measurements with past data and solves a convex projection problem to update the grid topology estimate $\theta_v(t)$ and water network parameter estimate $\theta_w(t)$. In the control decision layer, uncertainty is handled by introducing robust buffer terms k_v and k_w :

$$k_v = \hat{\eta}_v(t) + \rho_v \left(\frac{1}{\delta_v} + \|u_v\|_2 \right), \quad (34)$$

$$k_w = \hat{\eta}_w(t) + \rho_w \left(\frac{1}{\delta_w} + \|u_w\|_2 \right) \quad (35)$$

The design of these robust buffers ensures safe system operation even in the presence of parameter estimation errors. Defining the parameter errors as:

$$\Delta\theta_v = \|\theta_v^* - \hat{\theta}_v\|_{\Delta, \delta_v}, \quad \Delta\theta_w = \|\theta_w^* - \hat{\theta}_w\|_{\Delta, \delta_w} \quad (36)$$

where $\theta_v^* = (R^*, X^*, \eta_v^*)$ and $\theta_w^* = (\Phi^*, \eta_w^*)$ denote the true parameters. When the parameter errors satisfy $\Delta\theta_v \leq \rho_v$ and $\Delta\theta_w \leq \rho_w$, the robust controller guarantees system safety, i.e., $v(t+1) \in [\underline{v}, \bar{v}]$ and $h(t+1) \in [\underline{h}, \bar{h}]$. This holds for all disturbances satisfying $\|\omega_v(t)\|_\infty \leq \eta_v^*$ and $\|\omega_w(t)\|_\infty \leq \eta_w^*$.

The robust safety margins ρ_v and ρ_w quantify the maximum tolerable estimation errors in topology and parameters. Provided that the actual estimation errors do not exceed ρ_v and ρ_w , solving the joint water–electricity optimization problem using the NCBC-generated estimates within the robust predictive control framework yields optimal control actions that ensure system safety. For the PDN, the robust safety margin is given by:

$$\rho_v = \frac{\delta_v \epsilon_v}{1 + \delta_v \|\bar{u}_v - \underline{u}_v\|_2} \quad (37)$$

where $\epsilon_v > 0$ is the voltage safety margin constant; $\delta_v > 0$ is the weighting parameter that reflects the relative emphasis on disturbance estimation; $\bar{u}_v$ and $\underline{u}_v$ are the upper and lower bounds on the power grid control input u_v , representing the available control capability of the PDN. Similarly, the robust safety margin for the WDN is defined as:

$$\rho_w = \frac{\delta_w \epsilon_w}{1 + \delta_w \|\bar{u}_w - \underline{u}_w\|_2} \quad (38)$$

From the preceding theoretical analysis, finite bounds on safety violations can be established. For PDN, the cumulative number of voltage violations N_v over the entire operating period (i.e., the total number of time steps satisfying $v(t) \notin [\underline{v}, \bar{v}]$) is finite and satisfies:

$$N_v \leq \frac{2\gamma_v}{\rho_v} \text{diam}(\mathcal{X}_v \times [0, \bar{\eta}_v]) + 1 \quad (39)$$

Analogously, for WDN, the cumulative number of hydraulic head violations N_w over the entire operating period (i.e., the total number of time steps satisfying $h(t) \notin [\underline{h}, \bar{h}]$) satisfies:

$$N_w \leq \frac{2\gamma_w}{\rho_w} \text{diam}(\mathcal{X}_w \times [0, \bar{\eta}_w]) + 1 \quad (40)$$

In particular, the constant term 1 represents the potential number of violations during the system initialization. The theoretical foundation relies on the observation that each voltage or hydraulic head violation provides additional information about the true system, thereby reducing the uncertainty set and accelerating convergence toward the actual parameters. Given that the initial uncertainty is finite and the learning algorithm achieves a bounded competitive ratio, the total number of such violations must also be finite.

Although PDN and WDN are coupled through pump power, the error bounds are derived separately for each subsystem. This is because coupling effects are already captured within the respective disturbance terms $\omega_v(t)$ and $\omega_w(t)$. While pump power consumption increases nodal loads and may lead to more frequent voltage violations in the coupled system compared to an isolated grid, the total number of violations will not exceed the established bound N_v . Consequently, the overall safety of the water–electricity coupled systems can be bounded by the sum of the subsystem error counts, i.e., $N_{\text{total}} \leq N_v + N_w$. This result ensures that, even under simultaneous uncertainties in grid topology and hydraulic parameters, the system can recover to safe operating conditions within a finite number of steps, demonstrating the robustness of the proposed control framework.

5. Case study

This section presents comprehensive numerical simulations to validate the effectiveness of the proposed online learning-based robust voltage control strategy for water–electricity coupled systems under topology and parameter uncertainties.

5.1. Simulation environment and system parameters

The simulation is conducted on a coupled system consisting of a Southern California Edison (SCE) 56-bus PDN and a 15-node WDN. The simulations are implemented in Python 3.11 and run on a hardware platform equipped with an Intel(R) Core(TM) i7-2400H processor at 2.50 GHz over a 24-hour operational period. PDN and WDN exhibit significant differences in physical dynamics and equipment regulation characteristics. The grid side has fast variations at the second-level time scale, whereas the water side is not suitable for frequent regulation [37]. Therefore, a multi-time-scale modeling scheme is adopted: the PDN control step is set to 6 s, while the WDN control step is set to 1 h [21]. For the water–electricity coupled system, the key real-time control requirement is that each online computation must be completed within the PDN control interval. The average computation time per online simulation step is 1.95 s, which satisfies the requirement for real-time online control. The topology of the coupled water–electricity system is shown in Fig. 3.

The radial PDN contains 56 buses, with bus 0 as the substation. Buses 2, 4, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 19, 20, 23, 25, 26, 32 are equipped with PV inverters that provide reactive power regulation within $[u_q, \bar{u}_q]$. While PV inverters can jointly regulate active and reactive power, their unidirectional generation and capacity constraints limit operational flexibility. Therefore, energy storage systems are installed at buses 17, 29, and 41 to provide flexible active power regulation within $[u_p, \bar{u}_p]$. The voltage safety limits are set to $[\underline{v}, \bar{v}] = [0.95, 1.05]$ p.u., corresponding to squared voltage magnitudes of $[11.4^2, 12.6^2]$ kV². The nominal squared voltage target is $v_{\text{nom}} = (12 \text{ kV})^2$.

WDN is a 15-node tree-structured system with node 1 as the reservoir. It consists of 14 pipes, with flow rates restricted to $[0, 3240]$ m³/h. Three pumps are installed on pipes (1, 2), (4, 5), and (7, 8). Two storage tanks are located at nodes 10 and 13. Nodes 3, 11, and 15 serve as water demand nodes. The pressure head is constrained to $[\underline{h}, \bar{h}] = [20, 1000]$ m.

Regarding the algorithmic parameters, the weights are tuned to balance safety and optimality. Based on the weighting principles described in Section 3.3, the objective-function weights are set as $\alpha_v = 1$, $\alpha_p = 0.5$, $\alpha_q = 0.5$, $\alpha_w = 0.5$, $\alpha_T = 0.5$ and $\lambda = 0.5$. The slack-variable penalty weights are set as $\beta_v = 10$ and $\beta_w = 10$. Since all objective terms are normalized, the weighting coefficients only reflect the relative priorities among different control objectives rather than absolute physical scales. $\bar{\eta}_v$ is the disturbance upper bound determined from historical data, which is obtained by adding a 5% safety margin to the true-topology calculation result $\max_t \|\omega_v(t)\|_\infty = 8.63$, yielding $\bar{\eta}_v = 9.06$. $\epsilon_v = 0.1$ is the offline safety margin, which is used to further reserve a safety margin beyond the maximum disturbance. $\delta_v = 20$ is used to balance the model-parameter estimation error and the disturbance-upper-bound estimation error. Similarly, the water-side values are determined as $\bar{\eta}_w = 6.00$,

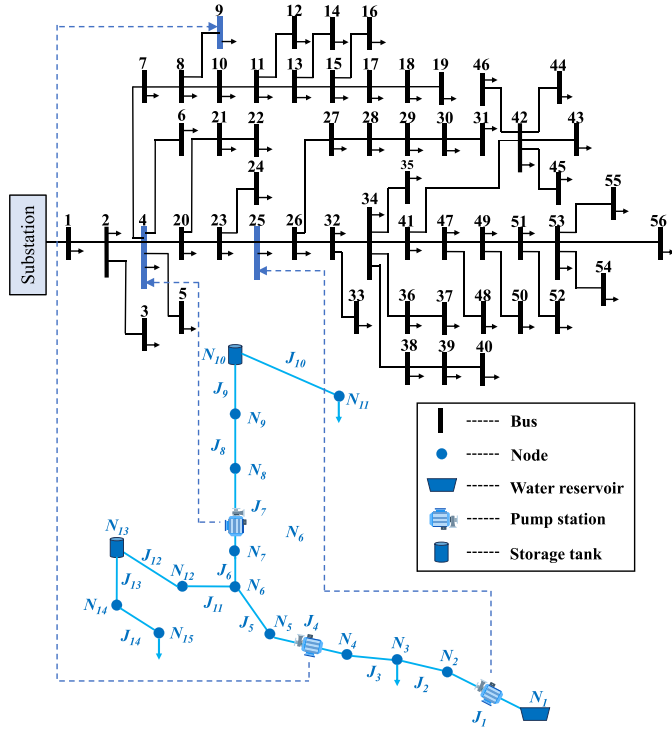


Fig. 3. Topology of the water-electricity coupled systems.

$\epsilon_w = 2.0$, and $\delta_w = 20$. Then, the robust radii ρ_v and ρ_w are calculated by (37) and (38). In the baseline simulations, the active- and reactive-power regulation limits are set as $\underline{u}_p = \underline{u}_q = -0.2$, $\bar{u}_p = \bar{u}_q = 0.2$.

5.2. Simulation results and analysis

First, to further validate the effectiveness of the proposed method under different levels of prior information, the voltage regulation performance is compared in a linear environment under three scenarios: fully unknown, partially known, and fully known PDN topology information. The comparative results are shown in Fig. 4 and Table 2. To examine the adaptability of the proposed algorithm under the most unfavorable information condition, the WDN pipe parameters are assumed to be fully unknown in all three cases. In Fig. 4, the orange lines and buses represent known PDN topology information, the red buses indicate locations where voltage violations occur, and the blue buses denote pump connection buses. On the PDN side, the entire dispatch horizon consists of 14,400 time steps, with each step lasting 6 s. In Table 2, violation instants are defined as the number of PDN dispatch time steps at which the voltage magnitude of at least one bus satisfies $v_i(t) \notin [v, \bar{v}]$.

Under the fully unknown topology scenario, the controller initially lacks PDN connectivity and line-parameter information, while it must also handle pump power fluctuations caused by unknown WDN parameters. Therefore, the voltage regulation task is relatively challenging. In this case, voltage violations occur at 32 buses over a total of 1302 time instants. The maximum upper and lower voltage-violation magnitudes are 0.1139 kV and 0.1182 kV, respectively, and the average violation magnitude is 0.0322 kV. As shown in Fig. 4(a), even under the extreme condition where the information on both the electricity and water sides is fully unknown, the controller can still maintain voltage security during most operating periods. In addition, Fig. 4(a) illustrates the evolution of topology-matrix uncertainty during the learning process. Driven by the NCBC algorithm, both R Gap and X Gap exhibit an overall decreasing trend, indicating that the topology estimates continuously approach their true values. The metrics R Gap and X Gap quantify the differences between the estimated and true topology matrices, and are defined as

$$R \text{ Gap} = \|\hat{R} - R^*\|_{\Delta}, \quad X \text{ Gap} = \|\hat{X} - X^*\|_{\Delta}. \quad (41)$$

Fig. 4(b) presents the simulation results under partially known topology, where the topology and line parameters of buses 0–24 are assumed to be known. The availability of partial prior information significantly reduces the initial uncertainty set and accelerates the topology learning process. Compared with the fully unknown case, the number of buses with voltage violations decreases from 32 to 6. The maximum upper voltage-violation magnitude decreases from 0.1139 kV to 0.0464 kV, the maximum lower voltage-violation magnitude decreases from 0.1182 kV to 0.0668 kV, and the average violation magnitude decreases from 0.0322 kV to 0.0225 kV. Both the number of violated buses and the violation severity are substantially reduced, indicating that partial prior information can effectively narrow the spatial range and reduce the severity of voltage violations. In this case, violations are mainly concentrated at a few critical buses and specific disturbance periods rather than appearing as large-scale violations.

For the fully known topology scenario, as shown in Fig. 4(c), voltage violations are not completely eliminated, but their severity is further reduced. The results show that the total number of voltage-violation time instants decreases to 1160. The maximum upper and lower voltage-violation magnitudes decrease to 0.0294 kV and 0.0326 kV, respectively, and the average violation magnitude decreases to 0.0178 kV. These results demonstrate that accurate topology information can significantly weaken the adverse impact of model uncertainty on voltage control performance.

To verify the applicability of the proposed method in a nonlinear environment, additional simulations are carried out, and the results are shown in Fig. 5. The results indicate that, under the fully unknown, partially known, and fully known scenarios, nodal voltages can still be effectively regulated around the safe operating range, and no persistent large-magnitude voltage violations occur. Moreover, in the fully unknown case, the initial R Gap and X Gap are relatively large, indicating that the controller initially lacks accurate knowledge of the PDN. As a result, voltage fluctuations are more pronounced. In contrast, under the partially known case, the convergence processes of R Gap and X Gap become smoother, and the voltage fluctuation magnitude is further reduced. These results demonstrate that the robust buffer terms in RMPC can compensate for linearization errors, model mismatches, and unmodeled dynamics. Therefore, the controller designed based on the linear approximation model can still remain effective in a more realistic nonlinear system.

Furthermore, under the premise that the grid topology information is fully unknown, the hydraulic head variations of WDN nodes are examined under different prior information levels. The results are shown in Fig. 6 and Table 3. Instead, they are jointly determined by the node elevation and the minimum pressure head $\underline{h} = 20$ m. The elevations of nodes 7, 11, and 13 are 121 m, 156.65 m, and 124.13 m, respectively. Therefore, the corresponding lower hydraulic-head limits are 141.00 m, 176.65 m, and 144.13 m, respectively. On the WDN side, the entire dispatch horizon consists of 24 time steps, with each step lasting 1 h.

When the WDN parameters are fully unknown, the minimum hydraulic heads at nodes 7, 11, and 13 are 165.89 m, 201.02 m, and 148.63 m, respectively, and no hydraulic head violation occurs. Under parameter uncertainty, the controller tends to reserve larger hydraulic-head safety margins to offset the influence of parameter-estimation errors. When the WDN parameters are fully known, the minimum hydraulic heads at the three nodes are 153.88 m, 188.99 m, and 148.47 m, respectively. No hydraulic head violation occurs in this case either. However, compared with the fully unknown case, the nodal hydraulic heads are closer to the safety boundaries, and the average hydraulic-head margin decreases from 16.679 m to 12.341 m, as shown in Table 4. This result does not indicate a degradation of safety. Instead, it shows that when WDN parameters are more accurate, the controller no longer needs to reserve excessive safety margins for parameter-estimation errors. Therefore, unnecessary conservative regulation can be reduced

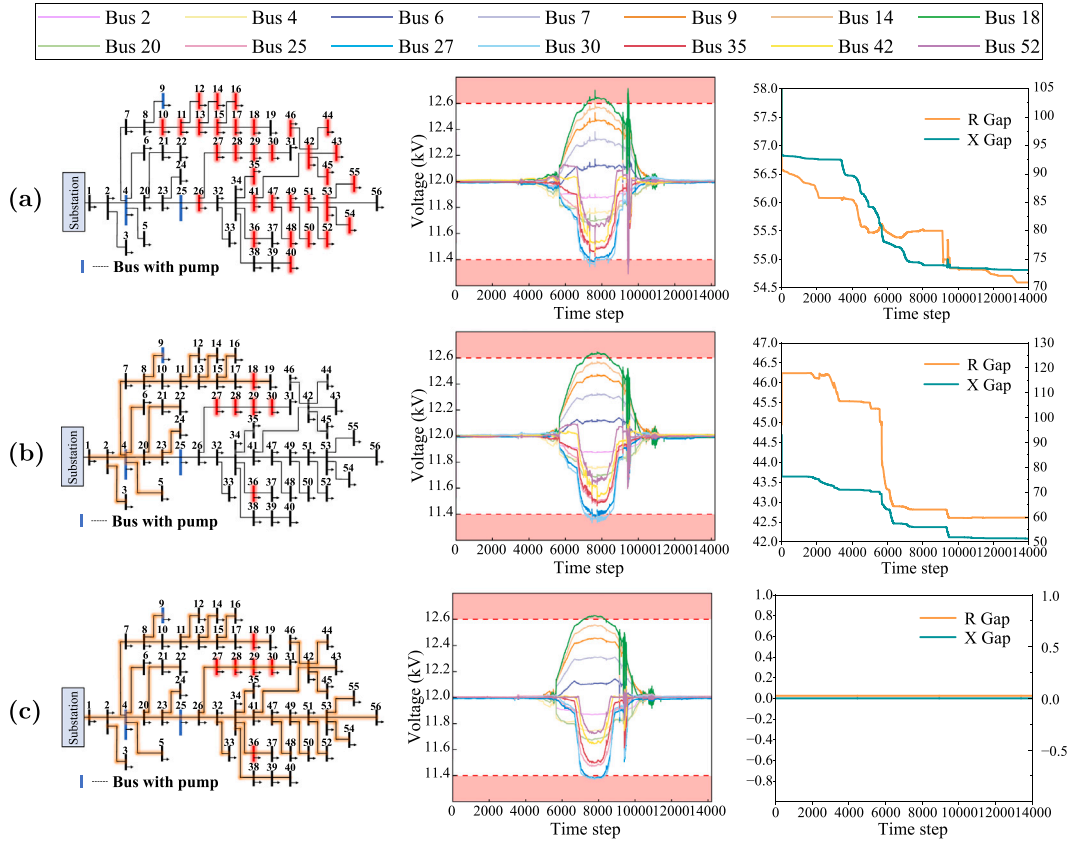


Fig. 4. Nodal voltage profiles and topology-learning convergence processes under different topology-knowledge levels in a linear environment.

Table 2
Voltage violation statistics under different prior information levels.

Metric	Fully unknown	Partially known	Fully known
Violation instants	1302	1286	1160
Violated buses	32	6	6
Max. overvoltage (kV)	0.1139	0.0464	0.0294
Max. undervoltage (kV)	0.1182	0.0668	0.0326
Avg. violation (kV)	0.0322	0.0225	0.0178

while hydraulic-head safety is still guaranteed, leading to a more precise control strategy. In addition, Table 4 shows that no hydraulic head violations occur in the entire WDN under either the fully unknown or fully known case, demonstrating that the proposed method still has strong adaptability even when the pipe resistance parameters of the WDN cannot be accurately obtained.

Fig. 7 illustrates the pump power variation and the impact of pump power modeling on PDN voltage control. Fig. 7(a) shows the pump power profile under unknown WDN parameters. The pump power varies significantly over the entire dispatch horizon, with an average power of 172.335 kW and a maximum total peak power of 220.764 kW. As a time-varying load, pump power directly affects nodal voltages. To further illustrate this impact, Fig. 7(b) presents the voltage control results without considering pump power under unknown grid topology. The results are compared with those obtained by considering pump power in Table 5.

Table 5 shows that under unknown grid topology, if pump power is ignored, the total number of voltage-violation time instants increases from 1302 to 1530, and the number of buses with voltage violations increases from 32 to 36. The maximum upper voltage-violation magnitude

increases from 0.1139 kV to 0.1223 kV, while the maximum lower voltage-violation magnitude increases significantly from 0.1182 kV to 0.1578 kV. This indicates that ignoring pump power causes the controller to underestimate the actual load level, making undervoltage more likely when pump-load fluctuations are large. Although the average violation magnitude becomes smaller when pump power is ignored, the violations occur more frequently, involve more buses, and exhibit larger extreme violation magnitudes. Therefore, the overall voltage security is worse. These results show that incorporating pump power into the voltage control model enables the controller to anticipate the influence of WDN operation on the PDN, effectively reducing the violation frequency and suppressing severe voltage deviations. This verifies the necessity of explicitly modeling water-electricity coupling.

Furthermore, to analyze the influence of control resources on voltage regulation performance, the active- and reactive-power control limits are increased from ± 0.2 to ± 0.3 , and simulations are conducted under the three prior-information scenarios. Fig. 8 shows that after the control capacity is increased, more regulation resources are available to the controller. Consequently, the voltage regulation ability of the system is significantly enhanced, the voltage fluctuation magnitude is further reduced, and no voltage violations occur during the entire dispatch horizon. This indicates that, in practical systems, increasing the regulation margins of controllable devices can help enhance the secure operation capability of water-electricity coupled systems under topology and parameter uncertainty.

Finally, to verify the advantages of the proposed unified water-electricity learning framework over conventional strategies, the proposed method is compared with the no control strategy and the two-stage control strategy. The results are shown in Fig. 9 and Table 6. The no control strategy applies no active regulation and is used to demonstrate

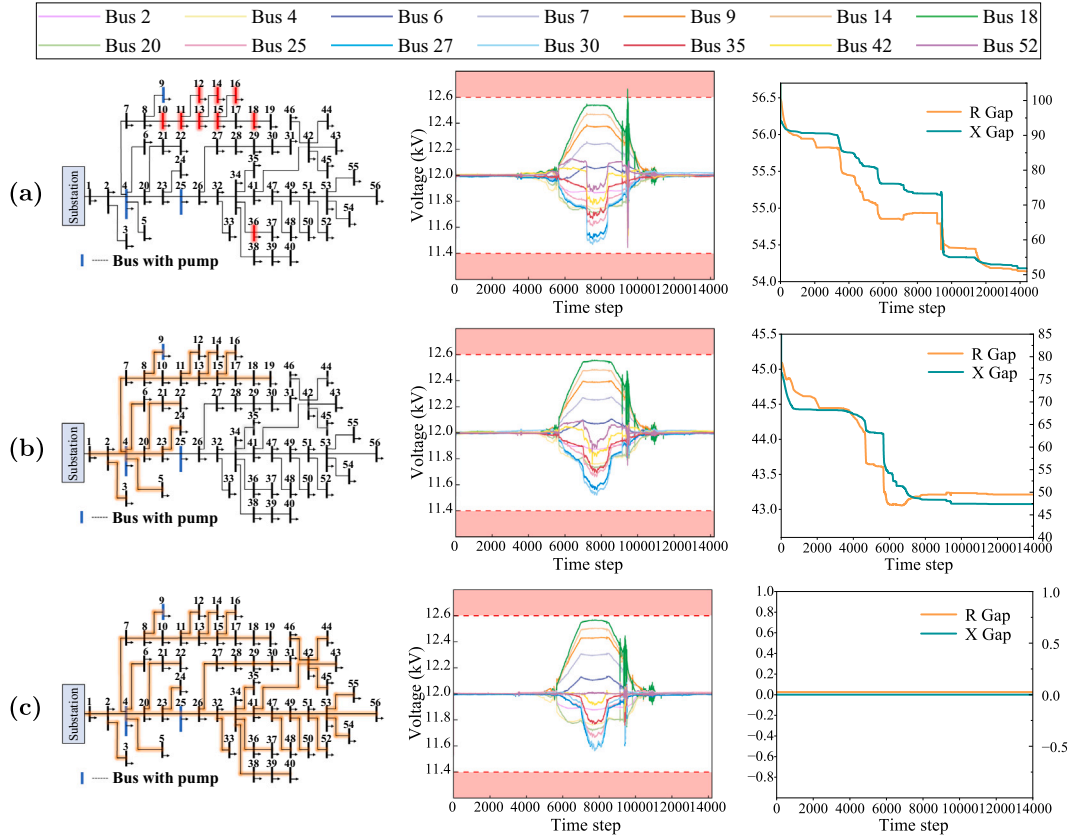


Fig. 5. Nodal voltage profiles and topology-learning convergence processes under different topology-knowledge levels in a nonlinear environment.

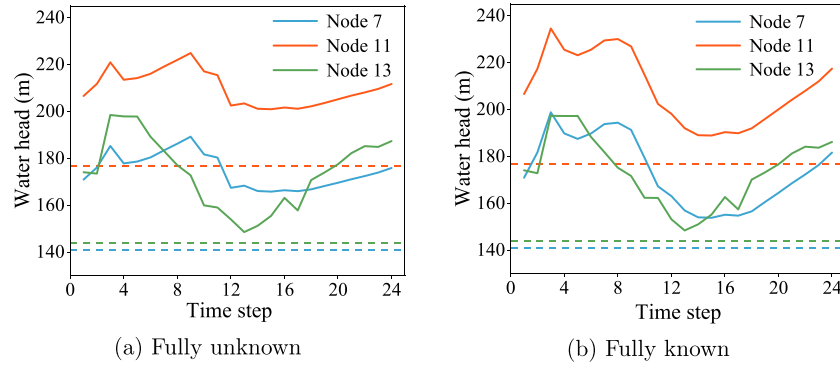


Fig. 6. Nodal hydraulic head variation under different levels of WDN parameter knowledge.

Table 3

Hydraulic head statistics under different prior information levels.

Scenario	Node	Lower limit (m)	Max. (m)	Min. (m)	Mean (m)
Fully unknown	7	141.00	189.31	165.89	174.60
	11	176.65	224.93	201.02	210.04
	13	144.13	198.51	148.63	174.02
Fully known	7	141.00	198.88	153.88	173.50
	11	176.65	234.56	188.99	208.98
	13	144.13	197.38	148.47	173.53

Table 4

Hydraulic head safety statistics under different prior information levels.

Metric	Fully unknown	Fully known
Number of nodes with head violations	0	0
Average hydraulic-head margin (m)	16.679	12.341

the necessity of voltage control. The two-stage control strategy first performs topology identification and then executes voltage control based on the identified model. Comparing this strategy with the proposed

method highlights the effectiveness of the unified learning-control framework.

Under the no control strategy, pump loads and external loads are directly imposed on the PDN. Due to the lack of active regulation capability, voltage violations are the most severe. Specifically, the total number of voltage-violation time instants reaches 10,976, and 44 buses experience voltage violations. The maximum upper and lower

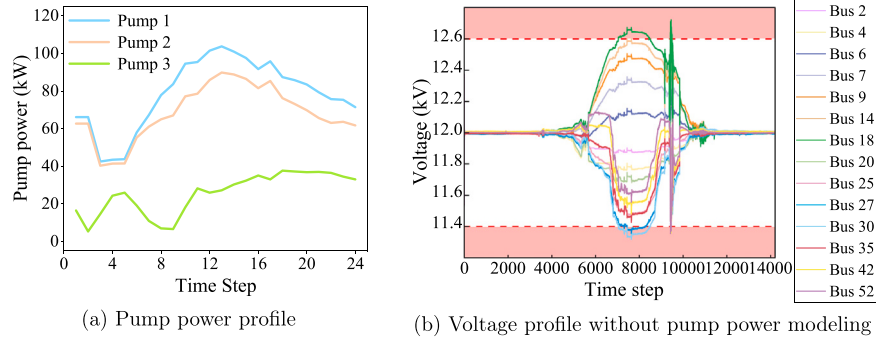


Fig. 7. Pump power variation and the impact of pump power modeling on voltage control.

Table 5
Impact of pump power modeling on voltage control performance.

Metric	With pump power	Without pump power
Violation instants	1302	1530
Violated buses	32	36
Max. overvoltage (kV)	0.1139	0.1223
Max. undervoltage (kV)	0.1182	0.1578
Avg. violation (kV)	0.0322	0.0189

voltage-violation magnitudes reach 0.6740 kV and 0.3081 kV, respectively, and the average violation magnitude is 0.1368 kV. This indicates that, under the combined influence of pump-load integration and renewable-generation fluctuations, the natural operation of the system alone cannot guarantee voltage security.

The two-stage control strategy performs topology identification before voltage control, and can reduce the number and magnitude of violations to some extent. However, inaccurate topology information may still lead to degraded control performance. Under the two-stage strategy, the total number of voltage-violation time instants is 4273, and 37 buses experience voltage violations. The maximum upper voltage-violation magnitude is 0.3659 kV, while the maximum lower voltage-violation magnitude reaches 0.3926 kV. The average violation magnitude is 0.1652 kV. Although the two-stage strategy performs better than the no control strategy, it is still significantly inferior to the proposed method. These results demonstrate that, compared with conventional control methods, the proposed method can maintain superior control performance under incomplete topology and parameter information in water-electricity coupled systems, highlighting its robustness and effectiveness.

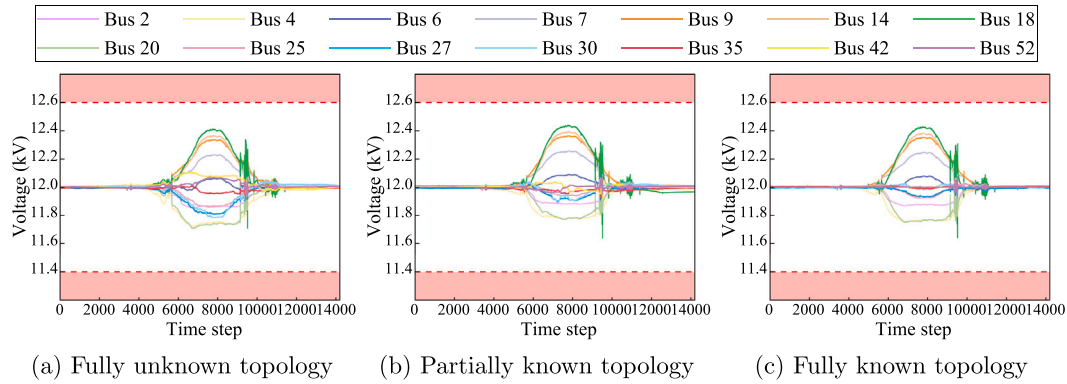


Fig. 8. Nodal voltage profiles under different topology-knowledge levels with enhanced control capabilities.

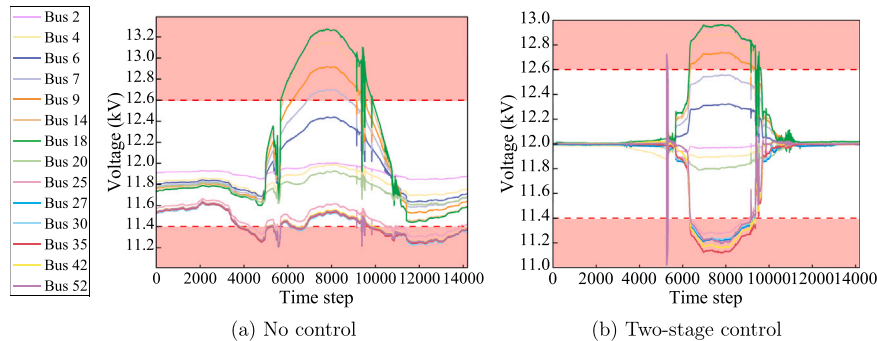


Fig. 9. Comparison of voltage regulation performance under different control strategies.

Table 6

Comparison of voltage regulation performance under different control strategies.

Metric	This work	No control	Two-stage
Violation instants	1302	10,976	4273
Violated buses	32	44	37
Max. overvoltage (kV)	0.1139	0.6740	0.3659
Max. undervoltage (kV)	0.1182	0.3081	0.3926
Avg. violation (kV)	0.0322	0.1368	0.1652

6. Conclusion

This paper proposes a robust voltage control strategy to address the challenges of incomplete power grid topology and water network parameter uncertainty in water–electricity coupled systems. A unified learning framework based on NCBC is established, enabling the joint on-line identification of grid topology and pipeline resistance coefficients relying solely on real-time operational data. Building upon this, a robust model predictive controller incorporating a dynamic robust buffer term is designed. Theoretical analysis provides provable safety guarantees by establishing finite upper bounds on the number of voltage and hydraulic head constraint violations, even under worst-case scenarios. Compared with conventional baseline strategies, the proposed approach significantly mitigates the risk of safety constraint violations. The simulation results further confirm the effectiveness of the proposed method under incomplete system information. Even in the extreme case where both the PDN topology and the WDN parameters are completely unknown, the proposed method largely maintains system-wide voltages within the prescribed safety limits. The average voltage–violation magnitude over the entire dispatch horizon is only 0.0322 kV, which is substantially lower than the 0.1368 kV and 0.1652 kV observed under the no control and two-stage control strategies, respectively. Moreover, increasing the available control resources noticeably improves the voltage regulation performance. For the WDN, no hydraulic-head violations are observed under either the fully unknown or fully known parameter setting, with average hydraulic-head safety margins of 16.679 m and 12.341 m, respectively. These results demonstrate the strong control performance and robustness of the proposed learning-control framework.

Although the proposed model yields promising results, its applicability is subject to several conditions. First, the method relies on sufficiently reliable communication and accessible measurements; delays, packet losses, or missing observations may hinder the contraction of NCBC feasible sets and compromise control performance. Second, scalability remains a concern, as computational burden and solution time may increase with system size, potentially restricting real-time deployment in large scale water–electricity coupled networks. Future work will be devoted to addressing issues such as incomplete communication and distributed or multi-agent implementation.

CCRediT authorship contribution statement

Jicheng Zhang: Writing – original draft, Validation, Methodology, Investigation. **Jiajia Chen:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Dechang Yang:** Writing – review & editing, Formal analysis. **Cong Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Derivation of pump power linearization

The active power consumption of a water pump is governed by the following fundamental hydraulic relationship:

$$P_{\text{pump}}(Q_{ij}) = \frac{\rho g}{\eta_p} h_{ij}^{\text{pump}} Q_{ij} \quad (\text{A.1})$$

where $P_{\text{pump}}(Q_{ij})$ denotes the active power consumed by the pump; ρ is the water density; g is the gravitational acceleration (typically 9.81 m/s²); η_p represents the pump efficiency; and Δh_{ij} is the pump head, defined as the hydraulic head difference between the outlet node j and the inlet node i .

Substituting the pump head-flow curve into the power equation and defining the constant coefficient $\zeta = \frac{\rho g}{\eta_p}$, we obtain:

$$P_{\text{pump}}(Q_{ij}) = \zeta \left(h_{0,ij} Q_{ij} - S_{ij} Q_{ij}^{\gamma+1} \right) \quad (\text{A.2})$$

Analysis of the second-order derivative reveals that $(P_{\text{pump}}(Q_{ij}))'' \leq 0$, indicating that the pump power function is concave. To incorporate this into a convex minimization framework, a linear upper bound is required to ensure feasibility. Since the tangent line of a concave function serves as a global overestimator, we employ the first-order Taylor expansion at the operating point Q_0 as the linear upper bound:

$$P_{\text{pump}}(Q_{ij}) \leq P_{\text{pump}}(Q_0) + \frac{d(P_{\text{pump}})}{dQ_{ij}} \Big|_{Q_{ij}=Q_0} (Q_{ij} - Q_0) \quad (\text{A.3})$$

By calculating the function value $P_{\text{pump}}(Q_0)$ and its derivative at Q_0 , and substituting them into the tangent equation, we derive the expanded inequality:

$$P_{\text{pump}}(Q_{ij}) \leq \zeta \left(h_{0,ij} Q_0 - S_{ij} Q_0^{\gamma+1} \right) + \zeta \left(h_{0,ij} - S_{ij}(\gamma+1) Q_0^\gamma \right) (Q_{ij} - Q_0) \quad (\text{A.4})$$

Rearranging the terms yields the final linearized upper bound:

$$P_{\text{pump}}(Q_{ij}) \leq \zeta \left[(h_{0,ij} - (\gamma+1) S_{ij} Q_0^\gamma) Q_{ij} + \gamma S_{ij} Q_0^{\gamma+1} \right] \quad (\text{A.5})$$

For simplicity, let $a_{ij}^p = \zeta [h_{0,ij} - S_{ij}(\gamma+1) Q_0^\gamma]$ denote the power-flow slope coefficient, representing the sensitivity of power consumption to flow variations. Furthermore, let $b_{ij}^p = \zeta S_{ij} \gamma Q_0^{\gamma+1}$ denote the constant power term at the specific operating point. The simplified linear constraint is thus expressed as:

$$P_{\text{pump}}(Q_{ij}) \leq a_{ij}^p Q_{ij} + b_{ij}^p \quad (\text{A.6})$$

Appendix B. Derivation of upper bound on constraint violations

This appendix formally presents the complete derivation of the robust buffer term and the maximum number of constraint violations discussed in the main text. The derivation relies on the key assumption of offline feasibility conditions stated in the main text, which guarantee that control measures exist to satisfy safety constraints even under the worst-case external disturbances. Specifically, for any $\Theta^* \in \mathcal{X}_v$ and $v_{\text{ext}} \in \mathcal{V}_{\text{ext}}$, there exists a control input $u_v \in [\underline{u}_v, \bar{u}_v]$ such that the resulting voltage v falls within a safe interval $[\underline{v} + \epsilon_v \mathbf{1}, \bar{v} - \epsilon_v \mathbf{1}]$ equipped with safety margins $\epsilon_v, \epsilon_w > 0$. The following derivation primarily focuses on the PDN side; the WDN side follows an identical logic.

The nodal voltage on the grid side at time step t can be expressed as:

$$v(t+1) = v(t) + \Theta^* u_v(t) + \omega_v(t) \quad (\text{B.1})$$

where $\omega_v(t)$ represents the unknown disturbance term satisfying $\|\omega_v(t)\|_\infty \leq \eta_v^*$. At time t , the NCBC algorithm provides a topology estimate $\hat{\Theta}$ and a disturbance upper bound estimate $\hat{\eta}_v$, yielding the voltage

prediction:

$$\hat{v}(t+1) = v(t) + \hat{\Theta}u_v(t) \quad (\text{B.2})$$

Subtracting the prediction from the actual voltage yields the prediction error:

$$v(t+1) - \hat{v}(t+1) = (\Theta^* - \hat{\Theta})u_v(t) + \omega_v(t) \quad (\text{B.3})$$

Applying the triangle inequality, we obtain:

$$\|v(t+1) - \hat{v}(t+1)\|_\infty \leq \|(\Theta^* - \hat{\Theta})u_v(t)\|_\infty + \|\omega_v(t)\|_\infty \quad (\text{B.4})$$

To unify the form, the right-hand side is expressed using a parameter radius. As defined in the main text, the uncertainty parameter and noise upper bound are measured using a weighted joint norm for the estimation error. From the definition of the joint norm, the noise upper bound component satisfies:

$$|\eta_v^* - \hat{\eta}_v| \leq \frac{1}{\delta_v} \|\theta^* - \hat{\theta}\|_{\Delta, \delta_v} \quad (\text{B.5})$$

When the estimation error falls within a neighborhood of radius ρ_v , i.e., $\|\theta^* - \hat{\theta}\|_{\Delta, \delta_v} \leq \rho_v$, we have:

$$\eta_v^* \leq \hat{\eta}_v + \frac{\rho_v}{\delta_v} \quad (\text{B.6})$$

Consequently,

$$\|\omega_v(t)\|_\infty \leq \eta_v^* \leq \hat{\eta}_v + \frac{\rho_v}{\delta_v} \quad (\text{B.7})$$

For the PDN block matrix $\Theta = [R \ X]$, where R and X are symmetric matrices, the block triangular norm is defined as

$$\|\Theta\|_\Delta^2 = \|R\|_\Delta^2 + \|X\|_\Delta^2. \quad (\text{B.8})$$

Let $E = \Theta^* - \hat{\Theta} = [E_R \ E_X]$ and $u_v(t) = [u'_p(t); u_q(t)]$. By the triangle inequality and the Cauchy-Schwarz inequality, we have

$$\begin{aligned} \|Eu_v(t)\|_\infty &= \|E_R u'_p(t) + E_X u_q(t)\|_\infty \\ &\leq \|E_R\|_\Delta \|u'_p(t)\|_2 + \|E_X\|_\Delta \|u_q(t)\|_2 \\ &\leq \|E\|_\Delta \|u_v(t)\|_2. \end{aligned} \quad (\text{B.9})$$

Therefore,

$$\|(\Theta^* - \hat{\Theta})u_v(t)\|_\infty \leq \|\Theta^* - \hat{\Theta}\|_\Delta \|u_v(t)\|_2 \leq \rho_v \|u_v(t)\|_2. \quad (\text{B.10})$$

Substituting the derived bounds into the prediction error equation, we obtain that when $\|\theta^* - \hat{\theta}\|_{\Delta, \delta_v} \leq \rho_v$:

$$\|v(t+1) - \hat{v}(t+1)\|_\infty \leq \hat{\eta}_v + \rho_v \left(\frac{1}{\delta_v} + \|u_v(t)\|_2 \right) \quad (\text{B.11})$$

Contracting the safety constraints inward by a radius k_v can offset the deviation caused by disturbances and the joint parameter estimation error. Therefore, the robust-buffer term for the PDN side is defined as:

$$k_v = \hat{\eta}_v + \rho_v \left(\frac{1}{\delta_v} + \|u_v\|_2 \right) \quad (\text{B.12})$$

This PDN robust buffer term is consistent with the equation in the main text. In the corresponding robust feasibility constraints, the buffer term similarly takes the form of k_v . Similarly, for the WDN side, we obtain:

$$k_w = \hat{\eta}_w + \rho_w \left(\frac{1}{\delta_w} + \|u_w\|_2 \right) \quad (\text{B.13})$$

Next, we derive the robust safety margin ρ_v . By the offline feasibility margin assumption, there exists a feasible control such that the system

falls within a contracted safety interval with an additional margin ϵ_v under the worst-case external disturbance bound $\bar{\eta}_v$. The online robust control employs a contraction radius k_v . To ensure that the online contraction does not exceed the maximum contraction allowed by the offline margin, a sufficient condition is:

$$k_v \leq \bar{\eta}_v + \epsilon_v \quad (\text{B.14})$$

Additionally, from $\theta \in \mathcal{X}_v \times [0, \bar{\eta}_v]$, it is known that the online estimate satisfies $\hat{\eta}_v \leq \bar{\eta}_v$. Substituting the buffer term expression and upper bounding $\hat{\eta}_v$ with $\bar{\eta}_v$, we obtain that satisfying the following is sufficient:

$$\bar{\eta}_v + \rho_v \left(\frac{1}{\delta_v} + \|u_v\|_2 \right) \leq \bar{\eta}_v + \epsilon_v \quad (\text{B.15})$$

which simplifies to ensure the online robust constraint is feasible. To make ρ_v independent of time t , it must hold for all feasible inputs within the control range $[u_v, \bar{u}_v]$. Thus, we transform the inequality using the bound of the input set:

$$\rho_v \left(\frac{1}{\delta_v} + \|\bar{u}_v - \underline{u}_v\|_2 \right) \leq \epsilon_v \quad (\text{B.16})$$

Rearranging the terms yields the final expression for the robust margin:

$$\rho_v \leq \frac{\delta_v \epsilon_v}{1 + \delta_v \|\bar{u}_v - \underline{u}_v\|_2} \quad (\text{B.17})$$

Similarly, for WDN side, we have:

$$\rho_w \leq \frac{\delta_w \epsilon_w}{1 + \delta_w \|\bar{u}_w - \underline{u}_w\|_2} \quad (\text{B.18})$$

Finally, we derive the upper bound on the number of constraint violations. This paper adopts a framework combining NCBC and robust predictive control. The core upper bound on violation counts relies on the property that when the estimation error $\|\theta^* - \hat{\theta}\|_{\Delta, \delta_v} \leq \rho_v$, the system is guaranteed to be safe. Replacing the generic robust margin ρ with ρ_v in the competitive ratio theorem, we obtain the upper bound on the number of grid voltage violations:

$$N_v \leq \frac{2\gamma_v}{\rho_v} \text{diam}(\mathcal{X}_v \times [0, \bar{\eta}_v]) + 1 \quad (\text{B.19})$$

Similarly for the WDN side, replacing ρ with ρ_w , we obtain the upper bound on the number of WDN hydraulic head violations:

$$N_w \leq \frac{2\gamma_w}{\rho_w} \text{diam}(\mathcal{X}_w \times [0, \bar{\eta}_w]) + 1 \quad (\text{B.20})$$

Data availability

Data will be made available upon request.

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